

| BIF5-SWEN3                            |                                                                                | Semester-Project/Code-Review Rating Matrix Final |          |                                |  |
|---------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------|----------|--------------------------------|--|
|                                       |                                                                                |                                                  | Team:    | Group O                        |  |
|                                       |                                                                                |                                                  | Student: | Gu Dan, Liu Zikun              |  |
|                                       |                                                                                |                                                  | Date:    | 12/1/2026                      |  |
|                                       |                                                                                |                                                  |          | <a href="#">Dann1223/SWEN3</a> |  |
| Title                                 | Description                                                                    | Max.                                             | Points   | Notes                          |  |
| <b>Functional-Requirements (20%):</b> |                                                                                |                                                  |          |                                |  |
| Use Cases                             | Understanding and Implementation of use-cases                                  | 5                                                | 5        |                                |  |
| REST API                              | Implementation of service-layers, converters, service-agents                   | 5                                                | 5        |                                |  |
| Web-Frontend                          | UX and completeness (functionallity) of web frontend                           | 5                                                | 4        |                                |  |
| additional Use Case                   |                                                                                | 5                                                | 3        |                                |  |
|                                       |                                                                                |                                                  |          |                                |  |
| <b>Non-Functional-Requirements:</b>   |                                                                                |                                                  |          |                                |  |
| Queues                                | Implementation of async-communication between components                       | 4                                                | 4        |                                |  |
| Logging                               | Appropriate logging implemented in all components/layers                       | 2                                                | 2        |                                |  |
| Validation                            | Validation in all layers and components                                        | 2                                                | 2        |                                |  |
| Stability Patterns                    | Exception handling, layer-based exceptions, stability patterns implemented     | 2                                                | 1        |                                |  |
| Unit Tests                            | Implementation of Unit-Tests with Mocking and appropriate coverage             | 4                                                | 2        |                                |  |
| Integration Tests                     | End2End Tests of use-cases and critical paths                                  | 4                                                | 1        |                                |  |
| Clean-Code                            | SOLID principles, clean-code for high code quality used                        | 2                                                | 2        |                                |  |
|                                       |                                                                                |                                                  |          |                                |  |
| <b>Software Architecture:</b>         |                                                                                |                                                  |          |                                |  |
| Packaging                             | Packaging and project-setup with containers (docker-compose, configs correct)  | 10                                               | 10       |                                |  |
| Loose Coupling                        | Implementation of interfaces, lose coupling                                    | 2                                                | 2        |                                |  |
| Mapper                                | Mapping between layers, usage of mapping framework                             | 2                                                | 2        |                                |  |
| Dependency Injection                  | DI implemented, DI framework used                                              | 2                                                | 2        |                                |  |
| DAL                                   | Implementation of persistence with ORM and Repository-Pattern (evtl. Unit-Of-W | 2                                                | 1.5      |                                |  |
| BL                                    | Implementation of BL, entities, workflow, facade                               | 2                                                | 1        |                                |  |
|                                       |                                                                                |                                                  |          |                                |  |
| <b>Software Development Workflow:</b> |                                                                                |                                                  |          |                                |  |
| GitFlow                               | GitFlow based; branches used; pull-requests used                               | 5                                                | 2        |                                |  |
| Issue Tracking                        | form of issue-tracking, kanban-board used and actual                           | 5                                                | 2        |                                |  |
| CI/CD-pipelines                       | some sort of CI/CD automation used (e.g. GitHub Action), for test, linting,..  | 5                                                | 4        |                                |  |

|  |                                                                                                                                                                                                                                                                                                               |                                                                                                      |            |             |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------|-------------|--|
|  | Documentation                                                                                                                                                                                                                                                                                                 | critical aspects of the solution are documented (diagrams, README.md)                                | 5          | 5           |  |
|  |                                                                                                                                                                                                                                                                                                               |                                                                                                      |            |             |  |
|  | <b>Code-Review (questions):</b>                                                                                                                                                                                                                                                                               |                                                                                                      |            |             |  |
|  | Knowledge                                                                                                                                                                                                                                                                                                     | Knowledge of the WHY/HOW/WHERE in the code.<br>Able to explain the background (theoretical) concepts | 20         | 10          |  |
|  |                                                                                                                                                                                                                                                                                                               |                                                                                                      |            |             |  |
|  |                                                                                                                                                                                                                                                                                                               | <b>Total (in %):</b>                                                                                 | <b>100</b> | <b>72.5</b> |  |
|  |                                                                                                                                                                                                                                                                                                               |                                                                                                      |            |             |  |
|  | <b>Remarks:</b><br>- The score here is in percent and will be scaled to the max-points of the code-review in Moodle<br>e.g. Mid-Term Code-Review 100% → 25 points; Final Code-Review 100% → 40 points<br>- You will get part score per item, if the task is not implemented complete or source quality is low |                                                                                                      |            |             |  |
|  |                                                                                                                                                                                                                                                                                                               |                                                                                                      |            |             |  |
|  |                                                                                                                                                                                                                                                                                                               |                                                                                                      |            |             |  |
|  |                                                                                                                                                                                                                                                                                                               |                                                                                                      |            |             |  |
|  |                                                                                                                                                                                                                                                                                                               |                                                                                                      |            |             |  |